Project Initialization and Planning Phase

Date	25 September 2026
Team ID	SWTID-2026-3161
Project Title	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to secure sensitive records in ServiceNow using script-controlled ACLs, improving data security and access control. It tackles the limits of role-only access rules, promising safer data, lower risk and better compliance. Key features include field-value based record restriction and thorough testing.

Project Overview	
Objective	The primary objective is to secure sensitive records in ServiceNow by using a script-controlled Access Control List (ACL) that allows or denies access to a record based on the value of a field in that record.
Scope	The project covers creating a custom table, users, groups and roles, and writing ACL condition scripts that restrict read and write access to records. It also covers field-level restriction, impersonation testing and performance testing of the ACL scripts.
Problem Statement	
Description	Standard role-based ACLs cannot restrict access using record-specific field values (for example, a record marked Confidential or assigned to another department). As a result, unauthorized users can view or modify records they should not see.
Impact	Solving this will protect sensitive data, prevent unauthorized changes, and improve security and compliance. It also reduces the risk of data leaks and builds trust in the platform.
Proposed Solution	
Approach	Use ServiceNow's script-controlled ACLs. A server-side JavaScript condition script reads the field value of the record (for example, current.field_name) and its context, and returns true or false to grant or deny access. The rules are verified by impersonating users with different roles.
Key Features	- Script-controlled ACL that restricts record access based on a field value. - Separate read and write ACLs for authorized and unauthorized users. - Field-level ACLs to hide sensitive fields. - Testing with user impersonation and a performance test of the ACL scripts.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard laptop/PC, Intel i5 (4 cores); no GPU needed (ServiceNow runs on the cloud)
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	256 GB SSD (project data is stored on the ServiceNow instance)
Software		
Frameworks	Platform / frameworks	ServiceNow Platform (Personal Developer Instance)
Libraries	Additional libraries / APIs	Glide APIs (GlideRecord, GlideSystem), server-side JavaScript
Development Environment	IDE	ServiceNow Studio / Script Editor, Chrome browser
Data		
Data	Source, size, format	Manually created test records on a custom table (about 20-30 records), stored in the ServiceNow database